

Colony Tracking Output Data Fields

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Overview

- The main colony tracking output is a parquet table under tracks/, then restitched into stitched/per_track/. The table is long-form: each row is one bodypoint for one tracked ant at one frame.
- Raw chunk files use chunk-local Frame values. Stitched per-track files use global block Frame values and add source_file, the chunk parquet that contributed each row.
- Coordinates are panorama coordinates in pixels. Downstream analyses usually convert pixels to millimeters with mm_per_px, commonly 0.016.

Track Parquet Fields

- Frame - Frame index. In tracks/*.parquet this is chunk-local. In stitched/per_track/*.parquet this is the global block frame after chunk offsetting.
- TrackID - Persistent ant identity, usually the ArUco tag ID. If ArUco is temporarily absent, the same TrackID can be continued by SLEAP or motion continuity.
- Bodypoint - SLEAP bodypoint index. Bodypoint names are not stored in the parquet. Bodypoint 0 is the default anchor used by speed and position analyses.
- X, Y - Panorama-space SLEAP coordinate for this specific bodypoint, in pixels. Because rows are bodypoint-level, the same ant and frame can have multiple X/Y rows.
- TrackX, TrackY - Tracking anchor coordinate for the ant in that frame, repeated on every bodypoint row. Usually this is the ArUco position when available; otherwise it is the SLEAP anchor or continuity position.
- ArucoX, ArucoY - Panorama-space ArUco marker coordinate assigned to this track/frame. Missing when no ArUco marker was assigned.
- SlepAnchorX, SlepAnchorY - Panorama-space SLEAP anchor bodypoint coordinate, usually bodypoint 0. Missing when the row is from ArUco-only continuation.
- CameraID - Preferred source camera for the row: ArUco camera if available, otherwise SLEAP camera.
- ArucoCam - Camera index that produced the ArUco detection. Missing values may appear as NaN when read with pandas.
- SlepCam - Camera index that produced the SLEAP anchor or instance. Missing values may appear as NaN when read with pandas.
- source_file - Present only in stitched/per_track/*.parquet. The source chunk parquet filename that contributed the row.

Parquet Metadata

- num_frames - Stored in parquet schema metadata for chunk and stitched outputs. It is the total frame span represented by that file or stitched group, including frames where a particular track may not be observed.
- The parquet schema declares Frame, TrackID, Bodypoint, CameraID, ArucoCam, and SlepCam as integer fields. When loaded into pandas, camera fields with missing values may be represented as floating columns with NaN.

Interaction Output

- Interaction files are separate parquets under interactions/. They contain directed antenna-to-body contacts, not bodypoint-level skeleton rows.
- Frame - Frame index within that interaction chunk file.
- antenna_track_id - TrackID of the ant whose antenna bodypoints are near another ant. The default antenna bodypoint indices are 4, 5, 6, 7, 8, and 9.
- body_track_id - TrackID of the contacted ant.
- Rows are directed. A to B and B to A are separate rows if both directions are detected.

Derived Speed Output

- speed_mm_s.npy - Dense per-frame speed vector in mm/s. Index i corresponds to frame frame_min + i from speed_metadata.json.
- NaN values mean no valid speed estimate for that frame. Speed is computed from TrackX/TrackY after optional short-gap interpolation and smoothing.
- speed_metadata.json fields include track_path, track_name, track_id, speed_path, speed_units, frame_min, frame_max, n_frames, n_observed_frames, n_valid_speed_frames, fps, mm_per_px, x_col, y_col, bodypoint_filter, max_interp_gap_frames, smooth_sigma_frames, max_speed_mm_s, and dtype.

Derived Colony Presence Output

- colony_presence_i1.npy - Dense per-frame inside/outside-colony vector. Values are -1 for missing XY, 0 for outside colony, and 1 for inside colony.
- Index i corresponds to frame frame_min + i from colony_presence_metadata.json.
- colony_presence_metadata.json records the input track, frame span, observed and valid frame counts, inside/outside counts, mm_per_px, coordinate columns, bodypoint_filter, colony_boxes_mm, and dtype.

Derived Grid Occupancy Output

- grid_occupancy_f4.npy - 2D occupancy histogram, normalized as bin_count / n_detected_frames. Values are fractions of detected frames.
- grid_x_edges_mm.npy and grid_y_edges_mm.npy - Bin edges for the histogram in millimeters.
- grid_occupancy_metadata.json records track identity, side, output paths, histogram shape, frame span, mm_per_px, grid size/padding, panorama bounds, whether side widths were equalized, coordinate columns, bodypoint_filter, and detection/grid inclusion counts.

Derived Sleep Prediction Output

- sleep_predictions.parquet - Per-frame sleep classifier output. Fields are Frame, track_name, track_id, side, wake_probability, sleep_probability, predicted_label_value, and predicted_label.
- predicted_label_value uses 0 for wake and 1 for sleep. Dense predicted_sleep_i1.npy uses -1 for missing frame, 0 for wake, and 1 for sleep.
- sleep_probability_f4.npy and wake_probability_f4.npy - Dense probability vectors aligned by frame_min in sleep_prediction_metadata.json.
- sleep_bouts.parquet - Contiguous predicted bouts with frame_start, frame_end, predicted_label_value, predicted_label, n_frames, mean_sleep_probability, and median_sleep_probability.
- sleep_prediction_metadata.json records model path, feature mode, feature count, speed root, fps, mm_per_px, speed windows, dense output paths, frame span, predicted frame counts, sleep/wake counts, sleep fraction, and mean sleep probability.

Source Code Locations

- Track output schema: /home/sam-reiter/saionHome/AntsArray/tracking/core/tracking_utils.py
- Stitching and source_file/global Frame offsets: /home/sam-reiter/saionHome/AntsArray/tracking/stitch_tracks.py
- Interaction output schema: /home/sam-reiter/saionHome/AntsArray/tracking/colony/interaction_one_chunk.py
- Derived speed/presence/grid/sleep outputs: /home/sam-reiter/saionHome/AntsArray/analysis/